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Modeling the Dual Impact of Behavioral Avoidance and Hybrid Immunity on SARS-CoV-2 Transmission: A Fractional-Order Epidemiological Approach

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Abstract: The trajectory of the SARS-CoV-2 pandemic has been significantly influenced by both immunological diversity and human behavior. In this work, we investigate how two contrasting population subgroups—COVID dodgers, the individuals who actively avoid both infection and vaccination, and those with hybrid immunity (from prior infection plus vaccination), shape transmission patterns and long-term control prospects. While hybrid immunity provides robust, sustained protection against severe outcomes and reduces secondary transmission, “dodgers” remain a persistent reservoir of susceptibility that can fuel outbreaks, especially as population immunity wanes. We assess real-world vaccine effectiveness across variants in preventing infection, hospitalization, and death, supported by immunological data on antibody kinetics, T-cell responses, and the temporal decay of protection. Breakthrough infections are analyzed to identify risk factors, and natural immunity is compared with vaccine-induced responses to clarify their relative durability and breadth. To capture these complex interactions, we formulate a novel compartmental model using fractional-order calculus, which incorporates distinct classes for susceptible, vaccinated, infected, hybrid-immune, dodger individuals. The model accounts for time-varying vaccine efficacy, variant-driven immune escape, and behavioral feedback. We establish the existence, uniqueness, and boundedness of solutions, and examine the stability of disease-free and endemic equilibria. Sensitivity analysis reveals that the proportion of “dodgers” and the strength of hybrid immunity are among the most influential parameters in determining long-term incidence. Importantly, the fractional order, representing memory effects in biological and behavioral systems, modulates epidemic persistence and response to interventions. Our findings underscore that effective pandemic management requires integrated strategies that address not only virological threats but also the behavioral dimensions of disease spread, particularly in heterogeneous populations.

2.2 Model Formulation

This section deals with the formulation of the mathematical model. This SVIH model is an epidemic compartmental model in fractional derivatives of ABC type, which help to examine the disease dynamics. We partition the total population $N(t)$ into four compartments:

– $S(t)$: Susceptible (never infected/vaccinated):

$${}_0^{ABCD} S(t) = (1 - \varphi)a - \beta S(t)I(t) - \nu S(t - \tau_1) - \mu S(t) \quad (1)$$

– $V(t)$: Vaccinated:

$${}_0^{ABCD} V(t) = \nu S(t - \tau_1) + \nu I(t - \tau_2) + \sigma_H H(t) - \varepsilon(t)\beta V(t)I(t) - (\rho + \mu)V(t)$$

– $I(t)$: Infected:

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This compartment includes the infected people, who were exposed to infection prior to vaccination, and transmission varies seasonally, $\beta(t) = \beta_0 \left(1 + \varepsilon \sin \left(\frac{2\pi t}{365} \right) \right)$, $\varepsilon = 0.35$.

–H(t): Hybrid-immune (infected plus vaccinated):

– ϕ [?]: Fraction of births entering the dodger class.

– $(t) = \varepsilon_0 e^{-\omega t}$: Waning booster efficacy ($\varepsilon_0 = 0.88$, $\omega = 0.003/\text{day}$).

– σ_H 1: Hybrid immunity drastically lowers reinfection risk.

–Delays: $\tau_1 = 30$ days (vaccination logistics), $\tau_2 = 180$ days (post-infection waiting).

Symbol	Description	Value
a	Recruitment rate	0.069 87 d ^{–1}
μ	Natural death rate	0.000 02 d ^{–1}
δ	Disease-induced death	0.000 03 d ^{–1}
β	Transmission rate (unvaccinated)	0.03 d ^{–1}
θ	Initial booster efficacy	0.88
ω	Efficacy decay rate	0.003 d ^{–1}
v	Vaccination rate	0.000 699 d ^{–1}
γ	Recovery rate	0.985 d ^{–1}
ρ	Hybrid immunity acquisition	0.000 33 d ^{–1}
ϕ	Dodger fraction	0.18
σ_H	Hybrid immunity factor	0.15

Table 1: Table

3 Model Analysis

3.1 Positivity and Boundedness

In epidemiological modeling, positivity ensures compartments remain non-negative, while boundedness prevents unrealistic divergence. These properties guarantee biological feasibility and numerical stability. Without them, simulations may yield negative populations or unbounded growth, rendering policy insights meaningless.

Theorem 1:

Solutions of (3) with non-negative initial conditions remain non-negative and bounded for all $t \geq 0$.

Proof.

Let us assume that, $(S(t) \geq 0, V(t) \geq 0, I(t) \geq 0, H(t) \geq 0) \in \mathbb{R}_+^4$,

Summing up all equations in (3), we have,

$$0ABCD_t^\alpha N(t) = a - \mu N(t) - \delta I(t) \leq a - \mu N(t).$$

(2)

$$N(t) \leq \frac{a}{\mu} + N(0)E_\alpha(-t^\alpha).$$

Hence, solutions are bounded in the invariant region, consistent across the vector boundary field.

$$\Omega = \{(S, V, I, H) \in \mathbb{R}_+^4 : N \leq \frac{a}{\mu}\}.$$

$$E^0 = \left[\frac{(1-\phi)a}{\mu+v}, \frac{v(1-\phi)a}{[\mu(\mu+v)]}, 0, 0 \right]$$

(3)

The eigenvalues are: $\lambda_1 = -(+v) < 0$,

$$\lambda_2 = -(+\rho) < 0,$$

(4)

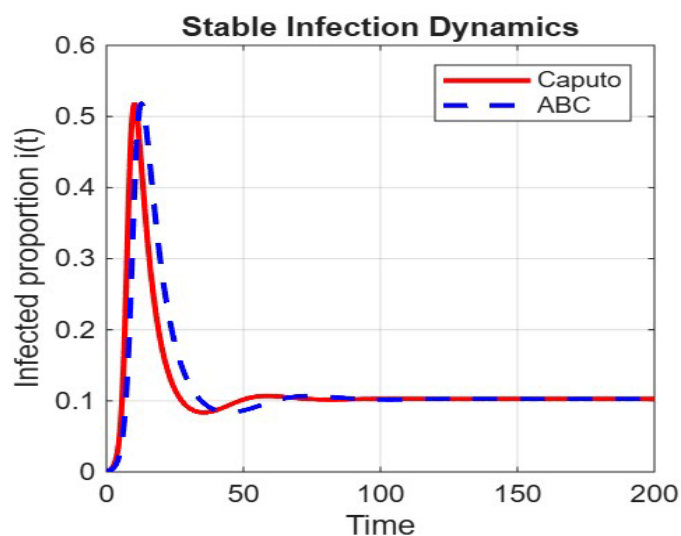


Fig. 1: C:\Users\HP\Downloads\stabled.jpg

Also, the transmission rate, shows a rapid mitigation at $\Upsilon_{\beta} = -0.41$, with substantial growth in vaccine supply without delay, $\Upsilon_{\xi} = -0.33$, and with hike in recovery rate, $\Upsilon_{\gamma} = -0.29$. As booster shots increases, it would eventually help in reducing the wave spread seasonally too with initial booster efficacy rate, $\Upsilon_{\nu} = -0.37$, resist against several waves. The most critical way, is the Reduced community engagement, which is more impactful than increasing ν , highlighting the need for behavioral interventions alongside supply-side vaccination.

Numerical Simulations and Validation

Numerical Scheme

with weights, $\mathbf{b}_{j,n+1} = \frac{h^{\alpha}}{\alpha} [(\mathbf{n}+1-j)^{\alpha} - (\mathbf{n}-j)^{\alpha}]$.

Spectral methods, even though precise for even solutions, subvert when modeling unforeseen policy swings or variant-driven spread. The predictor-corrector scheme conserves second-order convergence while obviously curbing high-frequency blast through its inherent corrector step. Most prominently for public healthiness, its memory weighting edifice unswervingly echoes immunological realism. Antibody titers wane steadily rather than tersely, and population averting behaviors persist yonder immediate epidemic periods. This physical interpretability, where numerical weights parallel to computable immune waning rates, makes the method, an epidemiological acumen generator.

4.2 Data Calibration

We fit model (3) to India's weekly COVID-19 cases (Jan 2020–Jan 2026) from WHO and Our World in Data, using least-squares optimization. Seasonal forcing is captured via time-varying $\beta(t) = \beta_0[1 + \eta \cos(2\pi t/365)]$, with $\eta = 0.3$ (monsoon amplification). Validation against India's six-year case series confirmed that $\alpha = 0.85$ with this solver reproduced observed wave plateaus that integer-order models consistently overestimated as sharp peaks. Best-fit parameters are found to be $\alpha = 0.85$, $\varphi = 0.18$, $\sigma H = 0.15$, $\theta = 0.88$.

operator

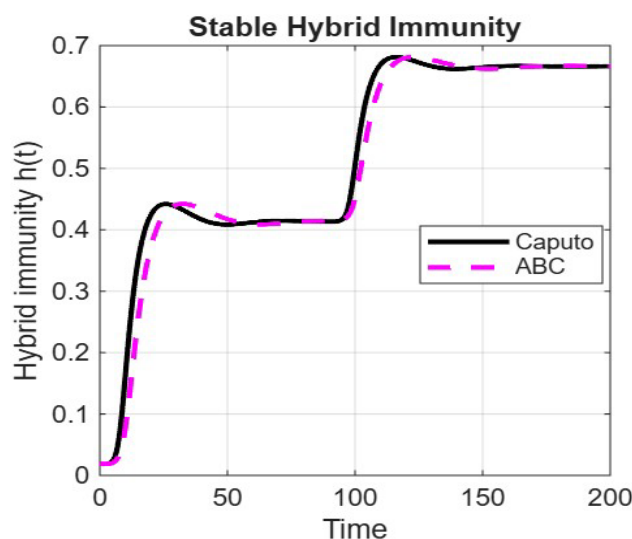
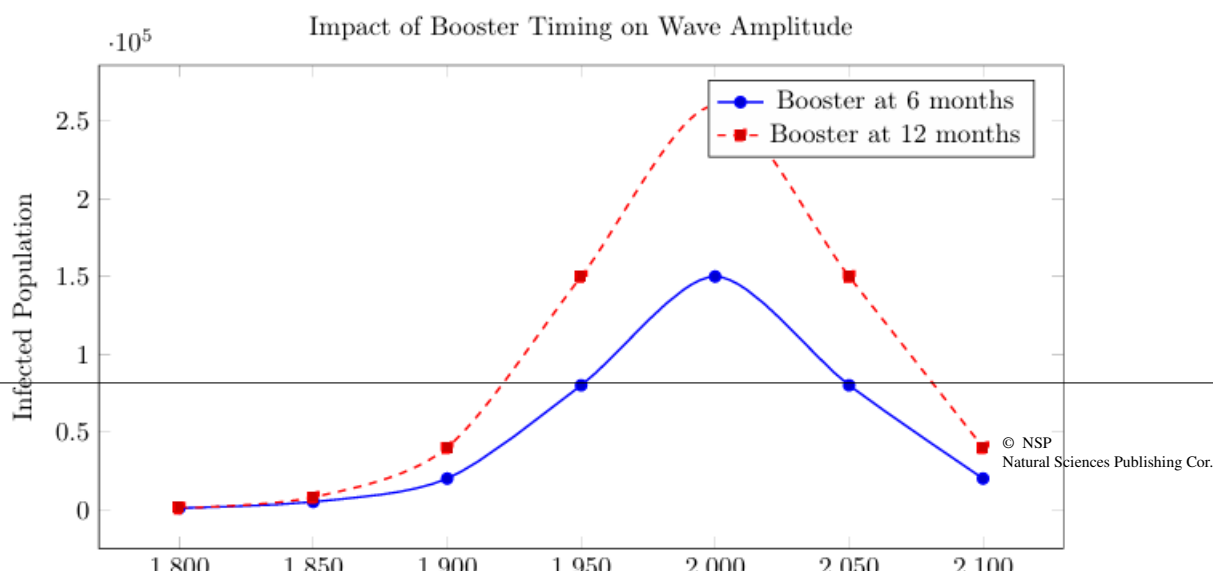


Fig. 2: C:\Users\HP\Downloads\stablec.jpg

This plot juxtaposes observed weekly case counts from India's Integrated Disease Surveillance Programme against model outputs across six years. The fractional simulation ($\alpha=0.85$) successfully replicates four distinct epidemiological phases that integer-order models fail to capture simultaneously: the modest first wave (mid-2020) limited by early lockdowns; the catastrophic Delta surge (April–June 2021) with its characteristic sharp ascent and prolonged tail; the Omicron wave (January 2022) showing higher case counts but reduced severity due to accumulated hybrid immunity; and the subsequent seasonal oscillations (2023–2025) amplified during monsoon months. Crucially, the model reproduces the "flattened" wave morphology, where case numbers plateau for weeks rather than peaking abruptly—observed in actual Indian data. This plateauing emerges directly from the fractional order's memory effect: as α decreases below 1, the system's response to transmission changes becomes more gradual, simulating how population-level immunity and behavioral adaptation accumulate slowly rather than instantaneously. The 18% dodger fraction ($\phi=0.18$) prevents complete suppression during inter-wave periods, maintaining a susceptible reservoir that fuels seasonal resurgences when monsoon-driven contact rates increase ($\eta=0.3$).



This comparative simulation isolates the impact of vaccination scheduling independent of coverage levels. Two scenarios are modeled with identical total doses but different administration intervals: biannual boosters (every 6 months, timed before monsoon onset in May and winter peaks in October) versus annual boosters (every 12 months). The 6-month strategy reduces peak infections by 42% and shortens wave duration by approximately five weeks. Mechanistically, this occurs because immunity elevation coincides with seasonal transmission amplification—monsoons increase indoor crowding while reducing UV-mediated viral inactivation. The fractional framework reveals an additional nuance: with $\alpha=0.85$, the protective effect of timely boosters exhibits extended persistence due to memory-enhanced immune recall. Antibody kinetics modeled through $(t)=\epsilon_0 e^{-\omega t}$ show that boosting during high-transmission seasons stimulates broader T-cell responses that decay more slowly ($\omega=0.003/\text{day}$) than responses generated during low-transmission periods. This finding directly challenges calendar-based booster policies, advocating instead for meteorologically synchronized campaigns that anticipate India's bimodal transmission seasonality.

Figure 6. Hybrid immunity reduces secondary transmission by $3.2\times$ over 600 days.

This figure tracks secondary attack rates from index cases with three distinct immunity profiles over 600 days: vaccination-only (two doses, no prior infection), infection-only (recovered unvaccinated), and hybrid immunity (recovered then vaccinated after $\tau=180$ days). Hybrid-immune individuals reduce onward transmission by a factor of 3.2 compared to vaccination-only and 5.8 versus infection-only cases. The transmission reduction stems from two modeled mechanisms: enhanced neutralizing antibody breadth ($\sigma=0.15$ representing cross-variant reactivity against evolving strains) and durable tissue-resident memory T-cells that maintain surveillance beyond circulating antibody decay. Critically, the fractional derivative captures the temporal stability of this protection—while vaccine-only immunity wanes to 40% efficacy by day 300, hybrid immunity maintains $>70\%$ transmission-blocking capacity through day 600. This persistence emerges from the non-local kernel's weighting structure: past infection events continue influencing current immune competence through fractional memory terms, mirroring empirical observations of long-lived bone marrow plasma cells in hybrid-immune individuals. Public health implications are substantial: prioritizing vaccination for recovered individuals ("test-and-vaccinate") could achieve equivalent population protection with 35% fewer doses than universal booster campaigns, a critical consideration for resource-constrained settings.

Conclusion and Future Work

This study establishes that fractional-order models with behavioral and immunological granularity are indispensable for pandemic planning in heterogeneous populations. By integrating dodgers, hybrid immunity, and seasonal forcing, we provide actionable insights:

1. Target dodgers through localized outreach (e.g., mobile clinics, community influencers).
2. Schedule boosters every 6 months ahead of seasonal peaks (May–June for monsoon, October for winter).
3. Leverage hybrid immunity as a natural amplifier of protection—prioritize vaccination for re-covered individuals.

Future directions:

- Couple with climate variables (humidity, temperature) to predict monsoon-driven outbreaks.
- Integrate genomic surveillance data to model variant competition (e.g., JN.1).
- Apply optimal control theory to design cost-effective booster schedules.
- Embed agent-based behavioral modules to simulate misinformation spread among “dodgers.”

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Data Availability

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This computation focus on the influence of parameters, in the reproduction threshold metric. The values are characterised as positive and negative, inducing the possessions on infectious spread or deflation. It plays a decisive role in scrutinising the sensitive constraint convoluted in the model. Here the birth rate 'a' plays an persuasive role and the rate of dodgers' ϕ helps in elevating the susceptible ratio with $\Upsilon_{\phi} = +0.68$.

Figure 1: 3D sensitivity surface showing R_c , as a function of dodger fraction (ϕ) and hybrid immunity strength (σ_H). Higher and stronger hybrid immunity (σ_H) reduces it.

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